

UDAYA SHANKAR

Senior Unity Developer | Game Developer

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🌐 udayashankar1.github.io/udayashankar.dev.portfolio 📍 Parnel, Maharashtra



SUMMARY

Passionate Senior Unity Developer with 5+ years of professional experience building games, simulations, and interactive applications using Unity 3D, Unreal Engine, and C#. Expertise in gameplay programming, performance optimization, cross-platform deployment (Android, iOS, WebGL, Windows), and scalable architecture using MVC, MVP, and event-driven design. Proven track record of delivering high-quality products, building CI/CD pipelines, developing custom editor tooling, and mentoring junior developers. Enthusiastic about AR/VR, real-time simulations, and emerging game development technologies.

EXPERIENCE

Senior Unity Developer

Immersivevision Technology

📅 12/2021 – 03/2026 📍 Pune, India

- Architected asynchronous content delivery pipelines using UniTask and Unity Addressable Asset System, improving runtime memory efficiency and reducing load overhead.
- Reduced draw calls and memory footprint by up to 35% through texture atlasing, GPU instancing, object pooling, and static/dynamic batching techniques.
- Authored Shader Graph and custom HLSL shaders for platform-optimised visual effects and stylised rendering, staying within mobile GPU performance budgets.
- Developed XR experiences targeting Meta Quest and mobile AR using Unity AR Foundation, XR Interaction Toolkit, and OpenXR standards.
- Developed custom Unity Editor extensions and inspector tools to automate repetitive production tasks, reducing team development time by approximately 30%.
- Applied software design patterns (MVC, MVP, State Machine, Observer) to engineer maintainable, decoupled, and testable gameplay systems.
- Engineered automated multi-platform build pipelines (Android, iOS, Windows, WebGL) using Unity Cloud Build and GitHub Actions CI/CD.
- Profiled and optimised runtime performance using Unity Profiler, Memory Profiler, and Frame Debugger, resolving CPU/GPU bottlenecks and maintaining stable framerates across all target platforms.
- Led structured code reviews and mentored junior developers on C# best practices, SOLID principles, and Unity architecture patterns.

Game Developer

Vr2ality

📅 02/2021 – 09/2021 📍 Pune, India

- Developed 10+ educational games for CBSE board curriculum in Unity, covering core academic subjects with interactive gameplay mechanics.
- Built diverse game genres including endless runners, survival, drag-and-drop, and collection games with dynamically updatable quizzes via Unity Asset Bundle.

EDUCATION

Bachelor in Computer Application

Seamedu school of Pro-Expressionism, Pune

📅 06/2018 – 08/2021 📍 Pune, India

SKILLS

Unity Engine (2D / 3D / HDRP / URP)

Advanced C# Programming & OOP

Unreal Engine 5 (Blueprints & C++)

C++ & Python Scripting

Addressable Asset System & Bundles

Async Programming (UniTask / Async-Await)

SOLID Principles & ScriptableObjects

Design Patterns (MVC, MVP, State Machine)

Multi-Platform Dev (Android, iOS, WebGL, PC)

CI/CD (Unity Cloud Build / GitHub Actions)

Performance Profiling & GPU Optimization

XR Development (AR / VR / Mixed Reality)

Custom Unity Editor Tooling

Unity DOTS / ECS / Burst Compiler

Shader Graph & HLSL / GLSL Shaders

AI-Assisted Development

PROJECTS

Car Simulation

Car Simulation project focused on realistic vehicle physics

- Developed a custom vehicle physics controller.
- Implemented acceleration, braking, steering, and drag calculations.
- Applied mathematical vehicle simulation principles.
- Designed configurable vehicle parameters using Scriptable Objects.
- Created debugging and telemetry tools for tuning.

Parkour Simulation

Parkour Simulation project involving complex character animations

- Implemented animation-driven movement using Animation Events and Target Matching.
- Built a character state machine for traversal transitions.
- Developed modular detection systems using raycasts and physics queries.
- Applied the Strategy Pattern for extensible traversal actions.